

The Volatility and Density Prediction Performance of Alternative GARCH Models

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ABSTRACT

This study compares the volatility and density prediction performance of alternative GARCH models with different conditional distribution specifications. The conditional residuals are specified as normal, skewed- t or compound Poisson (jump) distribution based upon a nonlinear and asymmetric GARCH (NGARCH) model framework. The empirical results for the S&P 500 and FTSE 100 index returns suggest that the jump model outperforms all other models in terms of both volatility forecasting and density prediction. Nevertheless, the superiority of the non-normal models is not always significant and diminished during the sample period on those occasions when volatility experiences an obvious structural change. Copyright © 2011 John Wiley & Sons, Ltd.

KEY WORDS GARCH; volatility forecasting; density prediction; skewed- t ; jump

INTRODUCTION

Volatility can be defined and interpreted in numerous ways, such as realized volatility, conditional volatility, and implied volatility. However, since it is a measure of price uncertainty, the forecasting of volatility is important for investment, derivatives valuation and risk management. While a volatility forecast is a number which provides only partial information about the distribution of the future price of an asset, a far more challenging task which can be of further benefit to derivatives traders and risk managers alike involves the use of market information to predict the entire distribution of the future asset price.

Density prediction is a different problem from volatility prediction but is related to it. When certain simulation or bootstrap techniques are applied, both tasks can be executed simultaneously with an asset price dynamic model. Although both volatility and density forecasts can be obtained from historical and/or option prices, the focus of this study is necessarily placed upon

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the historical models;¹ and indeed, amongst the various streams of the historical models, the generalized autoregressive conditional heteroskedasticity (GARCH)-type models, initiated by Engle (1982), extended by Bollerslev (1986) and Taylor (1986), and subsequently followed by many others, have proven to be the most successful models for describing and predicting asset price dynamics.

The success of the GARCH-type models comes as a result of their ability to capture many of the stylized facts of financial asset returns.² Although the conditional distribution of returns in a GARCH-type model is typically assumed to be normal, from their review of numerous studies Poon and Granger (2003) concluded that the standardized residuals from most GARCH-type models continued to display significant kurtosis. In other words, the higher-order features of the conditional distribution have tended to be completely ignored (Hansen, 1994); as a result, substantial return shocks may tend to be under-fitted by the normal models.

The literature aimed at providing a remedy for this problem has assumed at least two directions.³ The first approach was to directly model the stylized facts in the stochastic residuals with a flexible density (Bollerslev, 1987; Nelson, 1991; Hansen, 1994), while the second approach has attempted to capture the unusual and sudden changes in prices by including jump dynamics (Maheu and McCurdy, 2004; Duan *et al.*, 2006, 2007). To this end, it would seem natural to ask whether these efforts can always significantly improve the performance of the GARCH-type models in terms of certain economic applications, although they are of course necessary for superiority in an econometric sense. We therefore regard this study as contributing to the literature through its provision of a comprehensive comparison of alternative GARCH-type models (with different conditional distribution specifications) in terms of their volatility and density prediction performance.

As the asymmetric news impact has been well known for equity assets, the empirical comparison presented here is based upon the nonlinear and asymmetric GARCH (NGARCH) framework proposed by Engle and Ng (1993), with the standardized residuals being assumed to be normal, to investigate whether assuming the residuals to be (i) following skewed-*t* distribution (Hansen, 1994) or (ii) adapting to sudden return shocks by including jump dynamics (Duan *et al.*, 2006, 2007) can improve the precision of volatility and density prediction. Hence this study uses the traditional normal model (NGARCH-normal) as the benchmark to explore the incremental contribution of the non-normal specifications, comprising the skewed-*t* (NGARCH-skewed-*t*) and jump (NGARCH-jump) models, in volatility and density prediction for the S&P 500 and FTSE 100 indices.

¹ Poon and Granger (2003) provided a comprehensive review of volatility forecasting, while there has been extensive investigation into density prediction in many studies over recent years, such as Ritchey (1990), Madan and Milne (1994), Jackwerth and Rubinstein (1996), Melick and Thomas (1997), Malz (1996, 1997), Campa *et al.* (1998), Jondeau and Rockinger (2000), Rosenberg and Engle (2002), Bliss and Panigirtzoglou (2002), Liu *et al.* (2007), Wang (2009) and Shackleton *et al.* (2010), most of which were based upon the theoretical results for complete markets derived by Breeden and Litzenberger (1978); thus their focus was on option-implied densities.

² As indicated by Taylor (2005), there are at least four general properties that can be found in almost all sets of daily returns for financial assets, with the GARCH-type models having the capability of capturing them all. Firstly, the distribution of returns is non-normal; secondly, there is almost no correlation between returns for different days; thirdly, there is positive dependence between absolute/squared returns on nearby days; and fourthly, good news and bad news have different impacts on future volatility—the phenomenon referred to as asymmetry.

³ An alternative way of making the models more realistic is to increase the volatility memory length (Bollerslev and Mikkelsen, 1996); however, the information set used to form conditional volatility in this type of long-memory model differs from that of a short-memory model; therefore, no long-memory model is included in this study so as to avoid any unfair comparisons.

Based on 10,000 repeats of simulated prices and volatility values for many rolling samples,⁴ our empirical results for various forecasting horizons reveal that the return processes are better described by the non-normal specifications. Furthermore, the NGARCH-jump model is found to outperform the others in terms of both volatility forecasting and density prediction, although the achievement in the latter of these is less significant. Nevertheless, the advantages of these non-normal models on both applications are diminished for certain sample periods, in particular a period characterized by low, stable volatility. It may therefore be necessary to detect any structural changes in volatility, and to adapt to these when employing any of the historical models for time series modeling and forecasting, such as the GARCH-type models. Otherwise, a more complicated model that requires a much heavier computation load may not be superior to a simple one.

The remainder of this paper is organized as follows. The three volatility models employed in this study are described in detail in the next section, followed in the third section by presentation of the measures used to evaluate the volatility forecasting and density prediction performance of the various models. The fourth section presents the data used for our empirical analysis, with the analysis subsequently being presented in the fifth section along with a discussion of the results. Finally, the conclusions drawn from this study are presented in the sixth section.

EMPIRICAL MODELS

Many types of GARCH models, such as the EGARCH (Nelson, 1991), NGARCH (Engle and Ng, 1993) and GJR (Glosten *et al.*, 1993) models, are capable of capturing the asymmetric news impact that has been well documented for equity returns. As the focus of this study is on the distributional specifications of residuals (rather than the dynamic specifications of conditional variance) and the jump model we follow (Duan *et al.*, 2006, 2007) is specified for the NGARCH model, we simply use the NGARCH-normal model of Engle and Ng (1993) as the benchmark to investigate whether the NGARCH-skewed-*t* and the NGARCH-jump models can improve the precision of volatility and density prediction.⁵ This section provides details of the specifications and estimation procedures of the alternative models in a nested framework.⁶

Let $\{r_t\}$ denote the daily return process, and I_{t-1} be the information set comprising of all of the previous returns. The generalized NGARCH process is defined by

$$r_t = \alpha_t + h_t^{1/2} J_t \quad (1)$$

where $J_t \sim D(\mu_J, \sigma_J, \theta_0)$, and

⁴ Miguel and Olave (1999) and Pascual *et al.* (2006) use bootstrap approaches to predict volatility and density simultaneously. In particular, Pascual *et al.* (2006) incorporate parameter uncertainty and overcome the need for distributional assumptions. However, the procedure requires re-estimating parameters for every replication and thus using this approach will be extremely time-consuming for a complicated model like NGARCH-jump because it usually takes hours to finish one estimation procedure. The analysis here is therefore based on the results produced by conventional simulation techniques. Nonetheless, to check the robustness of our empirical findings, the results obtained from the approaches of Miguel and Olave (1999) and Pascual *et al.* (2006) will be compared with those generated by the conventional simulation procedure.

⁵ We also use the GJR framework for our empirical comparison and find that the results are very similar to those under the NGARCH framework. However, the levels of log-likelihood produced by the NGARCH-type models are higher than those obtained under the GJR-type models.

⁶ It is very important to have similar models when doing this kind of comparison.

$$h_t = \beta_0 + \beta_1 h_{t-1} + \beta_2 h_{t-1} \left(\frac{J_{t-1} - \mu_J}{\sigma_J} - c \right)^2 \quad (2)$$

where the conditional mean α_t and the conditional variance h_t vary with the information set I_{t-1} ; the residuals J_t are assumed to follow a particular distribution D , with mean μ_J , standard deviation σ_J , and other additional parameters θ_0 describing the density; and the conditional variance with positive c captures the (leverage) effect of the impact of information asymmetry. Given different distributional specifications for J_t , the three models we would like to explore are therefore obtained.

Model 1: the NGARCH-normal model

If J_t is assumed to follow standard normal distribution (i.e. $\mu_J=0$, $\sigma_J=1$ and θ_0 is an empty set), the NGARCH-normal model is thus obtained.

Model 2: the NGARCH-skewed- t model

Hansen (1994) proposed the skewed- t distribution, which, being very general, could simultaneously allow for both flexible skewness and kurtosis.⁷ In the NGARCH-skewed- t model, J_t is assumed to follow the standardized skewed- t distribution, with $\mu_J=0$, $\sigma_J=1$ and $\theta_0=[\eta, \kappa]$ respectively controlling for kurtosis and skewness of the density, which is given as

$$g(J_t|\eta, \kappa) = \begin{cases} bc \left(1 + \frac{1}{\eta-2} \left(\frac{bJ_t+a}{1-\kappa} \right)^2 \right)^{-(\eta+1)/2}, & J_t < -a/b \\ bc \left(1 + \frac{1}{\eta-2} \left(\frac{bJ_t+a}{1+\kappa} \right)^2 \right)^{-(\eta+1)/2}, & J_t \geq -a/b \end{cases} \quad (3)$$

where $2 < \eta < \infty$, and $-1 < \kappa < 1$. The constants a , b , and c are given by $a = 4\kappa c \left(\frac{\eta-2}{\eta-1} \right)$, $b = \sqrt{1 + 3\kappa^2 - a^2}$, and $c = \frac{\Gamma((\eta+1)/2)}{\sqrt{\pi(\eta-2)}\Gamma(\eta/2)}$.

Model 3: the NGARCH-jump model

As defined in Duan *et al.* (2006, 2007), the NGARCH-jump model is given by

$$J_t = z_t^{(0)} + \sum_{j=1}^{N_t} z_t^{(j)} \quad (4)$$

$\mu_J = \lambda\mu$ and $\sigma_J = \sqrt{1 + \lambda(\mu^2 + \gamma^2)}$.

Namely

$$h_t = \beta_0 + \beta_1 h_{t-1} + \beta_2 h_{t-1} \left(\frac{J_{t-1} - \lambda\mu}{\sqrt{1 + \lambda(\mu^2 + \gamma^2)}} - c \right)^2 \quad (5)$$

where $z_t^{(0)} \sim N(0, 1)$, $z_t^{(j)} \sim N(\mu, \gamma^2)$, for $j=1, 2, \dots$, and $N_t \sim \text{Poisson}(\lambda)$.

⁷ Following the Bollerslev (1987) proposition of standardized t distribution, Nelson (1991) went on to suggest a 'generalized error distribution' (GED), and while both of these do use an additional parameter to capture kurtosis, they are, nevertheless, still symmetrically distributed.

The compound Poisson is employed to handle the arrival of unusual news events which have an impact on returns.⁸ The jump size $Z_t^{(j)}$ is normally distributed with mean μ and variance γ^2 , while the number of jumps N_t is an invisible variable of non-negative integral with the arrival intensity λ being non-negative. It should be noted that unless $\lambda = 0$, the mean (variance) of J_t will not be 0 (1), but instead $\lambda\mu$ and $1 + \lambda(\mu^2 + \gamma^2)$, respectively. The normalized $J_{t-1}((J_{t-1} - \lambda\mu)/\sqrt{1 + \lambda(\mu^2 + \gamma^2)})$ in the last term of equation (5) serves to make this equation comparable to the NGARCH-normal model, which typically uses a random variable with mean 0 and variance 1 (Duan *et al.*, 2007). Intuitively, the existence of frequent large positive or negative return shocks indicates $\lambda > 0$ and brings significant kurtosis to the conditional distribution. A higher frequency of negative shocks, as compared to positive shocks, implies $\mu < 0$ and left skewness.⁹ Obviously, specifying a non-normal distribution for, and including jump dynamics in, the standardized residuals, both lead to the same consequence: a much more flexible conditional distribution of returns.

The ‘maximum likelihood estimation’ (MLE) method is employed to estimate the parameters for all models. The conditional density of the returns $f(r_t|I_{t-1}, \theta)$ can be obtained by transforming the density of the standardized residuals Z_t . Given that h_t and Z_t are functions of subset $\theta_1 \subset \theta$ and that the density of Z_t is determined by the subset $\theta_2 \subset \theta$, where $\theta_1 \cup \theta_2 = \theta$, the log-likelihood function of r_t is

$$\log L(\theta|r_1, \dots, r_n) = \sum_{t=1}^n -\frac{1}{2} \log(h_t(\theta_1)) + \log(f(z_t(\theta_1)|\theta_2)) \quad (6)$$

The density of Z_t in the NGARCH-normal model is a standard normal; therefore, $\theta_2 = \emptyset$. The density of Z_t in the NGARCH-skewed- t model is a standardized skewed- t density; therefore, $\theta_2 = \{\eta, \kappa\}$. The conditional density of r_t in the NGARCH-jump model is, however, more complicated since it includes two random variables (i.e. $Z_t^{(0)}$ and $\sum_{j=1}^{N_t} z_t^{(j)}$); thus the conditional density of r_t in the NGARCH-jump model is given as¹⁰

$$f(r_t|I_{t-1}, \theta) = \sum_{k=0}^{\infty} \frac{\lambda^k \exp(-\lambda)}{k!} N(\alpha_t + kuh_t^{1/2}, h_t(1 + k\gamma^2)) \quad (7)$$

PERFORMANCE EVALUATION

As compared to the NGARCH-normal model, the performance of the non-normal models is evaluated in terms of the precision of the volatility forecasting and density prediction. Not only are we able to

⁸ To capture any unexpected and significant reductions or increases in price, Press (1967) introduced a basic compound events model, in which unusual news events followed the Poisson process and the size of such movements was assumed to be normal. This jump idea has since been employed in GARCH-type models by Vlaar and Palm (1993), Chan and Maheu (2002), Maheu and McCurdy (2004), and Duan *et al.* (2006, 2007).

⁹ If the arrival intensity is allowed to be time-variant, the conditional skewness and conditional kurtosis will also be time-varying; however, Duan *et al.* (2007) indicated that in their model the variability of the arrival intensity was trivial. We also estimate the model with the time-varying arrival intensity following the specification of Maheu and McCurdy (2004) and our results coincide with the indication of Duan *et al.* (2007).

¹⁰ As suggested by Maheu and McCurdy (2004) and Duan *et al.* (2007), infinity in equation (7) is proxied by 50, which is large enough to make the summation converge.

obtain a set of simulated returns from the simulation procedure, but also a set of simulated variances. Therefore, we can evaluate the performance of alternative models on both volatility and density prediction from the same simulation procedure.

Volatility forecasting

For a GARCH-type model, given I_t , the one-period forecast (h_{t+1}) will be immediately known, while the H -period forecast can be obtained either from a simulation procedure or by taking the conditional expectation of the sum of H -conditional variances. For consistency with the procedure used to predict density, the empirical analysis is based on the simulated results.¹¹

Although there is only one random term per unit of time in the NGARCH-normal and the NGARCH-skewed- t models, we nevertheless have to simulate three random terms in the NGARCH-jump model. Based on the simulated variance values $\{h_{t+1}, h_{t+2}, \dots, h_{t+H}\}$, the H -period simulated variance can be computed as $h_H = h_{t+1} + h_{t+2} + \dots + h_{t+H}$. By repeating the simulation procedure 10,000 times, we can obtain 10,000 simulated values of h_H . The H -period volatility forecast is the square root of their mean.

The realized volatility calculated under the Andersen *et al.* (2001) approach is used as the target for forecasting; specifically, the daily realized variance is defined as the sum of the squared 5-minute returns during the day, and thus the total realized volatility over H days is the square root of the sum of H daily realized variances.¹²

We not only use the ‘mean squared error’ (MSE) and ‘root mean squared error’ (RMSE) values, but also, as suggested by Diebold and Mariano (1995), employ Wilcoxon’s signed-rank test for evaluation of the forecasting performance of the alternative models. By rolling over a fixed-length sample period for parameter estimation m times, we have m forecasts for a particular horizon, H .

The MSE and RMSE are respectively defined as $\frac{1}{m} \sum_{j=1}^m (f_{j,H} - y_{j,H})^2$ and $\sqrt{\frac{1}{m} \sum_{j=1}^m \left(\frac{f_{j,H} - y_{j,H}}{y_{j,H}} \right)^2}$,

where $f_{j,H}$ and $y_{j,H}$ respectively denote the j th forecast and realized volatility values over H days. To further compare the forecasting power statistically, we use Wilcoxon’s signed-rank test for the null hypothesis that the median of forecasting errors, measured by $(f_{j,H} - y_{j,H})^2$ or $\left(\frac{f_{j,H} - y_{j,H}}{y_{j,H}} \right)^2$, for the tested model is not less than that of the benchmark model. The test statistic is

$$S = \sum_{t=1}^m I_+(d_t) \text{rank}(|d_t|) \quad (8)$$

where d_t is the forecasting error of the tested model minus that of the benchmark model at time t . $I_+(d_t) = 1$ if $d_t > 0$, otherwise $I_+(d_t) = 0$. Moreover, its standardized version is asymptotically standard normal:

$$S_a = \frac{S - \frac{m(m+1)}{4}}{\sqrt{\frac{m(m+1)(2m+1)}{24}}} \stackrel{a}{\sim} N(0, 1) \quad (9)$$

¹¹ The volatility forecasts generated from the latter approach will be compared later for the robustness check.

¹² According to Andersen *et al.* (2001), the 5-minute frequency is an appropriate choice due to the tradeoff between microstructure effects and measurement precision.

Density prediction

Based on the simulated returns $\{r_{t+1}, r_{t+2}, \dots, r_{t+H}\}$, the asset price at time $t + H$ can be computed as $S_{t+H} = S_t^* \exp(r_{t+1} + r_{t+2} + \dots + r_{t+H})$, where S_t^* is the known price at time t . By repeating the simulation procedure 10,000 times, we can obtain 10,000 simulated values of S_{t+H} , denoted by $S^{(i)}$. These 10,000 simulated values can then be used to estimate the density of S_{t+H} , $\hat{g}_{t+H}(x)$, approximated by the normal kernel. After rolling over a fixed-length sample for parameter estimation m times, we can obtain m number of H -period-ahead density forecasts. Given their corresponding realized prices, $S_{j,H}^*$ for $j = 1, 2, \dots, m$, the probability integral transformations (PITs) can be obtained as

$$u_{j,H} = \int_{-\infty}^{S_{j,H}^*} \hat{g}_{j,H}(x) dx = \frac{1}{10,000} \sum_{i=1}^{10,000} \Phi \left(\frac{S_{j,H}^* - S^{(i)}}{B} \right), \text{ for } j = 1, 2, \dots, m \quad (10)$$

where $\Phi(\cdot)$ denotes the standard normal cumulative distribution function, and the bandwidth B is set at $\hat{\sigma}/10,000^{0.2}$ with $\hat{\sigma}$ being the standard error of the 10,000 simulated values.¹³

According to Diebold *et al.* (1998), $\{u_{j,H}\}_{j=1}^m$ will be independently and identically uniformly distributed if $\{\hat{g}_{j,H}(x)\}_{j=1}^m$ coincides with the series of the true (but unobservable) densities. In order to simultaneously test for distribution fitness and independence, Berkowitz (2001) proposed a more powerful approach in which $u_{j,H}$ is further transformed with the inverse of the standard normal distribution function as

$$v_{j,H} = \Phi^{-1}(u_{j,H}), \text{ for } j = 1, 2, \dots, m \quad (11)$$

where the inverse of the standard normal distribution function is denoted by $\Phi^{-1}(\cdot)$.¹⁴ If the series $\{u_{j,H}\}_{j=1}^m$ is an i.i.d. uniform distribution, then $\{v_{j,H}\}_{j=1}^m$ is an i.i.d. $N(0,1)$; in other words, if $\{\hat{g}_{j,H}(x)\}_{j=1}^m$ coincides with the true densities, then $\{v_{j,H}\}_{j=1}^m$ will be i.i.d. and it will follow standard normal distribution.

Berkowitz (2001) implemented the test by running the following model for $v_{j,H}$:

$$v_{j,H} - \mu_v = \rho_v (v_{j-1,H} - \mu_v) + \varepsilon_{j-1,H} \quad (12)$$

The log-likelihood function of this model was denoted as $L(\mu_v, \sigma_v^2, \rho_v)$, which will be nothing more than $L(0,1,0)$ when $\{v_{j,H}\}_{j=1}^m$ is i.i.d. $N(0,1)$.

In the joint test, the LR-ratio test statistic

$$\text{LR3} = -2 (L(0, 1, 0) - L(\hat{\mu}_v, \hat{\sigma}_v^2, \hat{\rho}_v)) \quad (13)$$

follows a $\chi^2(3)$ distribution under the null hypothesis; however, rejection of the above test may arise from the autocorrelation of the data series due to overlapped forecasts. Berkowitz (2001) therefore tested separately for autocorrelation by examining the following likelihood ratio statistic:

$$\text{LR1} = -2 (L(\hat{\mu}_v, \hat{\sigma}_v^2, 0) - L(\hat{\mu}_v, \hat{\sigma}_v^2, \hat{\rho}_v)) \quad (14)$$

which follows a $\chi^2(1)$ distribution under the null hypothesis.

¹³ As suggested by Silverman (1986), it is acceptable to set the bandwidth at $\hat{\sigma}/10,000^{0.2}$ since the standardized returns have unit variance.

¹⁴ Many other tests, such as the Kolmogorov–Smirnov test and the Anderson–Darling test, can be employed to test for distribution fitness only.

If $\hat{g}_{j,H}(x)$ provides accurate density, we cannot reject the null of LR3 and LR1. Conversely, if LR3 is rejected, this leads to two circumstances. Firstly, if LR1 is also rejected, the rejection of LR3 may be due to the series autocorrelation of the forecasts and it will therefore prove difficult to make any decision. Secondly, if LR1 is not rejected, then we can conclude that the estimated density is inaccurate.

DATA

Our empirical investigation is conducted on the S&P 500 and the FTSE 100 indices for a sample period running from 1 May 1997 to 29 April 2005. The primary dataset comprises of the high-frequency (5-minute) levels of these two indices obtained from Olsen. The closing price of a single trading day is taken from the last intraday price of the day.

The single-period returns are defined by the logarithmic price change, and thus the multi-period returns are the sums of the single-period returns. While the intraday returns are used to compute realized volatility, the daily returns are used as the inputs for the GARCH-type models.

Consistent with the stylized facts for equity returns found in many of the prior studies such as Andersen *et al.* (2001) and Areal and Taylor (2002), both the S&P 500 and the FTSE 100 index returns are negatively skewed and leptokurtic. The Jarque–Bera statistics further suggest that the return distributions of both indices are non-Gaussian.

While there appears to be no high autocorrelation for the daily returns of the S&P 500 index, the autocorrelations at lags 2 and 3 are slightly negative for the FTSE 100 index (−0.05 and −0.1, respectively). The results of the likelihood ratio tests do not, however, provide support for any particular specification for the conditional means of the returns of both indices.¹⁵ As a result, the conditional means of the returns for both indices are assumed to be constant in the following analyses.

The daily realized volatility series of the S&P 500 and the FTSE 100 indices are plotted in Figure 1, from which it is clearly evident that volatility clustering is present in both indices, with the volatility having been relatively stable and low over recent years, from about April 2003. Since the occurrence of extreme values seems to have been less frequent during this period, it may be quite interesting to explore whether the advantage of a model is dependent upon the volatility structure. This issue will be examined later in this study.

EMPIRICAL RESULTS

We apply three models to the daily returns of the S&P 500 and the FTSE 100 indices, using the NGARCH-normal model as the benchmark. Our study aims to examine whether the non-normal models, motivated under different perspectives, can provide any improvement on either volatility or density prediction. Before doing so, we first briefly summarize the model estimation or fitting results.

The MLE procedure is used to estimate the parameters in the models, with the likelihood-ratio (LR) tests and the Akaike information criteria (AIC) being used to evaluate the performance of the

¹⁵ For example, for the FTSE 100 index returns, the log-likelihoods for the NGARCH-normal model with the conditional mean specified as AR(1), MA(1), and ARMA(1,1) are 6250.22, 6250.22 and 6251.88, respectively, which are only slightly higher than the likelihood from the model with a constant conditional mean (6250.21).

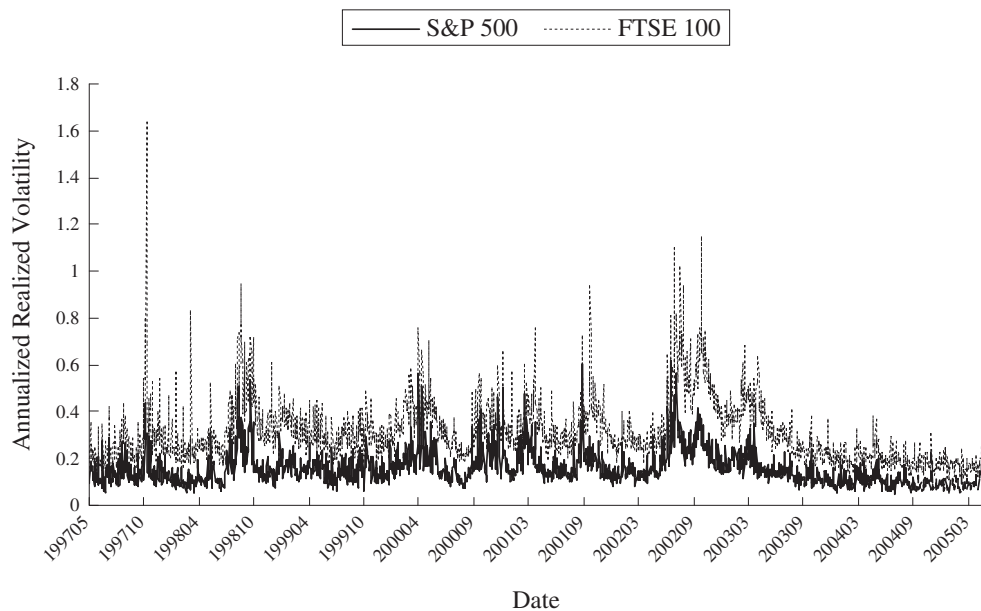


Figure 1. Annualized realized volatility. The figure shows the realized volatility series of the S&P 500 and the FTSE indices for the sample period from 1 May 1997 to 29 April 2005; all volatility values are annualized

three models in terms of model fitting. Consistent with previous studies such as Bollerslev (1987), Nelson (1991), Glosten *et al.* (1993) and Poon and Granger (2003), the estimates of parameters reveal many well-known properties of price dynamics of equity assets, including the inappropriateness of the normality assumption, the existence of volatility clustering and leverage effect, and the stationarity of the volatility processes. In general, the non-normal models have both higher log-likelihood and lower AIC values than the NGARCH-normal model, and indeed the LR tests also support the non-normal specifications of the conditional distribution.

The empirical results for volatility forecasting and density prediction are detailed in the following two subsections and followed by further discussion and robustness analysis.

Volatility forecasting

We take the observations for the last 3 years of the sample period as the out-of-sample period. The total volatility over the five horizons of 1, 5, 10, 20 and 30 days are forecast under the three alternative models, with the whole sample being exhausted by the use of the rolling procedure. The MSE and RMSE values based on the realized volatility and the standardized statistics of the Wilcoxon's signed-rank test based on the NGARCH-normal model are presented in Table I for the S&P 500 index, and in Table II for the FTSE 100 index.

The non-normal models generally outperform the normal model, with the NGARCH-jump model performing best of all, which is dependent on neither the assets nor the forecasting horizons. In particular, in terms of MSE (RMSE), the jump model improves volatility forecasting performance in the S&P 500 index by as much as 12.70% (7.52%) to 17.61% (9.74%). However, while the skewed-*t* model performs rather well in terms of model fitting, its performance in volatility forecasting is less

Table I. Volatility forecasting performance, S&P 500

Model	Horizon				
	1 day	5 days	10 days	20 days	30 days
<i>Panel A: MSE</i>					
NGARCH-normal	0.00244	0.00176	0.00197	0.00250	0.00298
NGARCH-skewed- <i>t</i>	0.00242 (−3.40)	0.00173 (−6.00)	0.00194 (−6.67)	0.00248 (−5.93)	0.00298 (−4.19)
NGARCH-jump	0.00213 (−16.71)	0.00145 (−18.94)	0.00164 (−19.78)	0.00213 (−18.72)	0.00260 (−17.28)
<i>Panel B: RMSE</i>					
NGARCH-normal	0.4347	0.3447	0.3554	0.4025	0.4461
NGARCH-skewed- <i>t</i>	0.4302 (−4.98)	0.3402 (−6.98)	0.3507 (−7.67)	0.3984 (−6.74)	0.4429 (−5.42)
NGARCH-jump	0.4020 (−19.20)	0.3122 (−20.63)	0.3207 (−21.42)	0.3655 (−21.17)	0.4084 (−20.30)

Note: Panel A presents the mean squared errors of volatility forecasting for the S&P 500 index returns, while panel B presents the root mean squared relative errors. The out-of-sample period runs from 1 May 2002 to 29 April 2005. Values in parentheses refer to the standardized statistics of Wilcoxon's signed-rank test for the improvement in forecasting based on the NGARCH-normal model.

Table II. Volatility forecasting performance, FTSE 100

Model	Horizon				
	1 day	5 days	10 days	20 days	30 days
<i>Panel A: MSE</i>					
NGARCH-normal	0.00311	0.00211	0.00230	0.00300	0.00356
NGARCH-skewed- <i>t</i>	0.00302 (−0.60)	0.00204 (−1.74)	0.00224 (−2.95)	0.00292 (−3.22)	0.00345 (−2.02)
NGARCH-jump	0.00291 (−8.59)	0.00191 (−9.69)	0.00209 (−10.37)	0.00274 (−11.35)	0.00325 (−10.74)
<i>Panel B: RMSE</i>					
NGARCH-normal	0.4213	0.3327	0.3501	0.4034	0.4548
NGARCH-skewed- <i>t</i>	0.4179 (−2.46)	0.3286 (−3.73)	0.3458 (−4.22)	0.4008 (−3.44)	0.4541 (−1.95)
NGARCH-jump	0.3867 (−12.55)	0.2991 (−14.23)	0.3140 (−15.03)	0.3634 (−16.28)	0.4126 (−16.06)

Note: Panel A presents the mean squared errors of volatility forecasting for the FTSE 100 index returns, while panel B presents the root mean squared relative errors. The out-of-sample period runs from 1 May 2002 to 29 April 2005. Values in parentheses refer to the standardized statistics of Wilcoxon's signed-rank test for the improvement in forecasting based on the NGARCH-normal model.

convincing. For both criteria under all horizons, the percentages of forecasting improvement made by the skewed-*t* model are less than 1.52%, with the one exception of 30-day forecasting in the S&P 500 index. Similar results can also be found for the FTSE 100 index.

These findings are also confirmed by the Wilcoxon's signed-rank test. Based on the NGARCH-normal model, the NGARCH-jump model has significantly lowered the forecasting errors for both

indices under all horizons with the significance levels less than 1%. However, the absolute values of the test statistics for the NGARCH-skewed- t model are much lower and sometimes the improvement in reducing forecasting errors is even insignificant under the significance level of 10% (1-day forecasting for the FTSE 100 index).

In summary, the NGARCH-jump model outperforms the other models, in terms of volatility forecasting performance, for both indices and for all forecasting horizons. This finding is consistent with that of Maheu and McCurdy (2004), although the jump component is specified differently.

Density prediction

Density prediction is also executed for the five horizons of 1, 5, 10, 20 and 30 days, with the Berkowitz (2001) approach being used to evaluate the performance of the alternative NGARCH models. Ten thousand simulated values are generated to form each density, and in order to avoid producing overlapping forecasts the forecasts are formed once per unit of horizon. For example, for the 5-day forecast, the forecasts are formed once every 5 days.¹⁶ The results of the Berkowitz (2001) tests are presented in Table III.

The results for the S&P 500 index presented in panel A of Table III suggest that all three models fail to provide accurate predictions for the 1-day densities, given that the LR3 statistics are rejected under the significance level of 5% but the LR1 statistics are not. All three models provide accurate estimated densities for other horizons, since all LR3 statistics are smaller than the critical value under the significance level of 5% (7.81), as are all LR1 statistics (3.84).

Thus, under the significance level of 5%, there appears to be no significant difference between the three models, in terms of their density prediction performance; however, under the significance level of 10%, the p -values suggest that the skewed- t and jump models provide good prediction for 5-day densities, whereas the normal model does not.

Furthermore, for all horizons (with the exception of the 20-day horizon), the LR3 statistics produced by the skewed- t and jump models are smaller than those produced by the normal model. According to Berkowitz (2001), a smaller (higher) LR3 statistic value (p -value) implies more accurate density prediction; hence it is reasonable to suggest that the non-normal models provide more accurate density prediction than the normal model, although such prediction superiority is not statistically significant.

The results for the FTSE 100 index presented in panel B of Table III are quite similar to those for the S&P 500 index. For the 10-, 20- and 30-day forecasts, all three models provide accurate estimated densities, given the rejection of the nulls of LR3 and LR1 under the significance level of 5%. However, under the significance level of 1%, only the NGARCH-jump model provides satisfactory density prediction for the 1-day forecasts. We therefore continue to assert that the non-normal models provide more accurate density prediction than the normal model. In addition, for all horizons, the LR3 statistics produced by the skewed- t and jump models are smaller than those produced by the normal model.

Thus it seems that capturing the higher-order features of conditional density can indeed improve the performance of density prediction, which is consistent with the finding of Wang (2009) using option-implied densities.

¹⁶ The total numbers of observations for the S&P 500 index are: 1-day forecasts (732); 5-day forecasts (147); 10-day forecasts (74); 20-day forecasts (37); and 30-day forecasts (25). The total numbers of observations for the FTSE 100 index are: 1-day forecasts (758); 5-day forecasts (152); 10-day forecasts (76); 20-day forecasts (38); and 30-day forecasts (26).

Table III. Density prediction evaluation

Horizon	NGARCH-normal		NGARCH-skewed- <i>t</i>		NGARCH-jump	
	LR3	LR1	LR3	LR1	LR3	LR1
<i>Panel A: S&P 500</i>						
1 day	10.63 (0.0139)	2.85 (0.0915)	8.38 (0.0387)	2.65 (0.1036)	8.17 (0.0426)	2.59 (0.1073)
5 days	6.36 (0.0953)	0.11 (0.7416)	6.03 (0.1099)	0.11 (0.7422)	6.11 (0.1062)	0.20 (0.6515)
10 days	4.11 (0.2498)	0.37 (0.5425)	3.71 (0.2944)	0.33 (0.5666)	3.91 (0.2710)	0.19 (0.6631)
20 days	3.12 (0.3732)	0.08 (0.7758)	2.82 (0.4202)	0.07 (0.7977)	3.55 (0.3137)	0.12 (0.7315)
30 days	3.44 (0.3287)	0.88 (0.3469)	3.27 (0.3521)	0.84 (0.3599)	3.31 (0.3468)	0.80 (0.3698)
<i>Panel B: FTSE 100</i>						
1 day	13.07 (0.0045)	4.63 (0.0314)	11.81 (0.0081)	4.64 (0.0312)	10.36 (0.0158)	5.04 (0.0248)
5 days	10.54 (0.0145)	0.47 (0.4910)	8.53 (0.0362)	0.30 (0.5814)	7.96 (0.0468)	0.15 (0.6951)
10 days	4.33 (0.2275)	0.27 (0.6032)	3.85 (0.2778)	0.23 (0.6309)	3.64 (0.3035)	0.06 (0.8079)
20 days	2.58 (0.4603)	0.31 (0.5765)	2.54 (0.4688)	0.38 (0.5378)	2.34 (0.5048)	0.21 (0.6433)
30 days	4.34 (0.2265)	1.28 (0.2582)	3.92 (0.2706)	1.13 (0.2884)	3.73 (0.2919)	0.82 (0.3640)

Note: The table presents the LR3 and LR1 statistics of Berkowitz (2001), with the LR3 statistics following a chi-square distribution with three degrees of freedom under the null hypothesis that the estimated densities coincide with the true densities, and the LR1 statistics following a chi-square distribution with one degree of freedom under the null hypothesis. Values in parentheses are *p*-values. The out-of-sample period runs from 1 May 2002 to 29 April 2005, with the forecast horizons covering 1-day, 5-day, 10-day, 20-day and 30-day periods; the forecasts for all horizons are non-overlapped. The decision criteria for the tests are: (i) if the nulls of both the LR3 and LR1 statistics are not rejected, then the estimated densities provide accurate forecasts; (ii) if the null of the LR3 statistic is rejected, but the null of the LR1 statistic is not, the estimated densities provide poor forecasts; and (iii) if the nulls of both the LR3 and the LR1 statistics are rejected, we are unable to draw any conclusion.

Further discussion

As shown in Figure 1, the volatility processes for both indices have been relatively stable and low over recent years; thus, in order to determine whether this phenomenon affects the adequacy of the non-normal densities, we divide the whole sample period evenly, and reinvestigate the volatility and density prediction performance of the three models for the two resultant subsamples.

Overall, the results for the second subsample, with stable, low volatility, differ significantly from the results for the whole sample and the first subsample, and the advantage of the non-normal models seems to be reduced substantially. Volatility is generally overestimated even by the NGARCH-jump model for the second subsample. In particular, the forecasting errors of the S&P 500 index for the second half of the sample period are about 1.6 times those for the first half. Similarly, for the second subsample, all three models provide poor density prediction for all horizons, with all of the LR3

statistics being larger than the critical values under the significance level of 5%, whereas the LR1 statistics are not.¹⁷

We may therefore suggest that the normal model may be appropriate for recent years, a period in which volatility became stably low, and the advantage of the non-normal models was quite limited. Namely, extreme events may be relatively unimportant for this period.¹⁸

In fact, the failure of all three models to provide good volatility and density prediction for the period with volatility becoming stably low is an inevitable shortcoming of virtually all historical models, and we may need to take into account structural changes in volatility when employing such historical models for volatility and density prediction.

Robustness analysis

It should be noted that not only can volatility forecasts be generated by the simulation procedure but also by the analytical solutions derived from taking the conditional expectations of the sum of H -conditional variances. Therefore, the simulation results of volatility forecasting from the simulation are compared with those from the analytical solutions, not only as a means of checking the robustness of our empirical results but also as a means of checking the quality of our simulation.

The MSEs and RMSEs of the alternative models for the two forecasting approaches show that the volatility forecasts from the simulation are almost identical to those from the analytical solutions. In particular, the maximum RMSE difference for the S&P 500 index is only 0.0003. Our empirical results and simulation procedures are therefore deemed to be reliable.

Although the bootstrap procedure proposed by Pascual *et al.* (2006) can integrate volatility forecasting and density prediction, overcome the need for the distributional assumption and incorporate parameter uncertainty, it is too computationally expensive for our study, especially for the NGARCH-jump model. Nonetheless, to further ensure the robustness of our empirical findings, we use 1000 repeats to check whether the distributions of price and volatility forecasts produced by our simulation procedure are different from those generated by the approaches of Pascual *et al.* (2006) as well as Miguel and Olave (1999).

The summary statistics of the distributions generated by alternative approaches show that the mean levels (of both volatility and price forecasts) do not obviously differ. In particular, the difference in volatility (price) for the S&P 500 index is smaller than 0.01 (5) in general. However, the difference between the 5% and 95% percentiles is generally wider for the distributions produced by the bootstrap procedure of Pascual *et al.* (2006), which is regarded as a natural outcome of incorporating parameter uncertainty. These robustness findings do not depend on the forecasting horizon.

In summary, our empirical findings are robust to the approaches used to generate volatility and price forecasts.

CONCLUSIONS

In an attempt to incorporate the higher-order features of the conditional distribution in the GARCH class models, we can either directly model the stylized facts in the stochastic residuals, with some

¹⁷ Since the sample period is divided into two equal parts, there are too few observations for 20- and 30-day forecasts; thus the forecasts for these two horizons are omitted here.

¹⁸ There is a rich literature on extreme-value-based tail estimation and density prediction. For example, see Longin (2000), McNeil and Frey (2000), Bali (2003, 2007) and Jalal and Rockinger (2008).

flexible densities, or make some attempt to capture the unusual changes in prices by including jump dynamics. In order to examine whether these adjustment approaches can provide any improvement in terms of volatility and density prediction performance as compared to the normal model, in this study we have implemented the NGARCH-normal, NGARCH-skewed- t and NGARCH-jump models for the S&P 500 and the FTSE 100 indices. We find that the non-normal models generally outperform the normal model in all respects, although their level of superiority in density prediction is not statistically significant.

However, we have also found that all of the advantages of the non-normal models over the normal model are substantially reduced when they are applied to any sample period that is characterized by stable and low volatility. In fact, a rather inevitable shortcoming of all the historical models is that the out-of-sample forecasts will tend to be seriously biased if there is any structural change in volatility. It may therefore be necessary to attempt to design a mechanism capable of adapting to structural changes when attempting to employ such historical models for volatility and density prediction. This is an issue that must be left to future research.

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